A Major Project Report
(Phase-I)

On

Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Model

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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Models", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Ms. Kirti**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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ABSTRACT

Given the increased use of solar photovoltaic (PV) energy in the power grid, there is a need for reliable power generation forecasts that will allow grid stability and proper load balancing. Nonetheless, production from solar energy is highly unpredictable since it greatly depends on complicated, non-linear meteorological parameters such as irradiance, temperature, and temperature of the module. This paper presents a solution in form of a machine learning model capable of generating short-term (hourly and daily) forecasts of AC power generation from the solar energy plants. Through accurate generation predictions, the proposed solution seeks to optimize the operation of the solar power plants, minimize the need for alternative power sources, and facilitate renewable energy grid integration. For the development of the algorithm, the data on the plant and weather sensors is extensively preprocessed and analyzed. For the purposes of this paper, forecasting is seen as a regression problem, where several algorithms including Linear Regression, Random Forest, Support Vector Regression (SVR), XGBoost and LSTM network algorithms are developed and trained.

PROBLEM DEFINITION

The fast integration of solar PV installations into power networks of today is restricted due to the intermittent nature of the weather, and a survey of the current state of the forecasting literature exposes important flaws in the available approaches. Advanced deep learning models, for instance, LSTM and other types of recurrent neural networks, are characterized by slow inference speed, absence of feature interpretation capability, and vulnerability to underfitting in case of training with small datasets, specific to particular plants. At the same time, conventional regression algorithms are often unable to detect sudden weather changes, handle sudden jumps in the power output, and perform unpredictably for different time scales. Besides, intensive ensemble methods require too much computing resources to become deployable at the edge.

Therefore, an important research challenge emerges to create an efficient but computationally light-weight solution. The main problem that this project seeks to solve is the design of an interpretable forecasting algorithm that would be able to efficiently map the input data of multiple sensors—solar irradiance, ambient temperature, and module temperature—to short-term AC power generation.

LITERATURE SURVEY

The rapid expansion of solar PV systems is critical for sustainable energy, but generation intermittency challenges grid stability. Traditional forecasting models struggle to capture complex, non-linear interactions between weather variables—like solar radiation and temperature—and power output. To address this, recent research has shifted toward data-driven machine learning. This literature survey explores the evolution of short-term solar forecasting, tracing the transition from conventional statistical methods to advanced algorithms like Random Forest, XGBoost, SVR, and LSTM. By reviewing prior studies on data preprocessing and performance evaluation (R^2, RMSE, MAE), this survey establishes the foundation for developing our multi-model predictive framework.

Serial Number	Year	Objectives	Methodology	Limitations
1	2024	To forecast short-term energy generation in a solar power plant.	Tested MLR, ARIMA, KNN, SVR, MLP, LSTM, Random Forest, XGBoost, and heterogeneous ensembles.	Nonlinear models like SVR worked much worse than simple linear models.
2	2024	To test predictive models on a vast dataset of Australian solar energy.	Evaluated ANN, Random Forest, Decision Tree, XGBoost, and LSTM.	Regression models often struggle to predict peak power values, necessitating classification approaches for high-generation spikes.
3	2019	To outline machine learning techniques used for short-term solar energy prediction.	Reviewed support vector regression, artificial neural networks, and hybrid models.	As a broad review paper, it lacks a unified empirical testing framework to benchmark the discussed algorithms identically.
4	2024	To solve consumption-production imbalances by uncovering intricate relationships between weather and energy production.	Applied XGBoost, LSTM, and Bi-LSTM models to a Kaggle dataset of climate parameters and solar panel output.	Complex deep learning architectures like LSTM are prone to underfitting on small datasets.
5	2025	To forecast solar energy in Zafarana, Egypt, utilizing region-specific	Proposed a hybrid model based on Random Forest and Gradient Boosting	The model's tuning to regional Egyptian climate data limits its immediate

		historical data.	(RFGB).	generalization to other geographical zones.
6	2025	To predict PV energy generation in an energy community across varying granularities.	Compared Random Forest for 24-hour ahead (30-min intervals) and XGBoost for 5-minute interval rooftop PV data.	The best-performing model varies depending on data granularity, preventing the use of a single universal algorithm.
7	2023	To optimize short-term solar energy forecasting using integrated computational intelligence.	Integrated Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU).	Recurrent neural networks demand significantly higher computational resources and training times compared to tree-based regressors.
8	2026	To predict short-term solar photovoltaic power generation on the DKASC Alice Springs dataset.	Compared 1D CNN, Bi-LSTM, Stacked LSTM, ANN, Linear Regression, SVR, and XGBoost, validated by the Diebold-Mariano test.	The high structural complexity of Stacked LSTM models reduces interpretability when analyzing individual feature importance.
9	2020	To evaluate short-term solar energy prediction by various machine learning algorithms.	Applied standard time-series machine learning regression algorithms to historical solar data.	Basic algorithms frequently fail to capture complex, non-linear spatiotemporal interactions during rapid weather fluctuations.
10	2023	To improve predictive robustness for solar power generation across diverse meteorological zones.	Employed stacking and voting ensemble learning methods combining bagging (Random Forest) and boosting.	Ensembling multiple heavy models introduces computational latency, making it less viable for instantaneous edge-device real-time monitoring.

OBJECTIVES

The primary objectives of this project are:

- To examine the shortcomings of the conventional approach to forecast systems in representing non-linear relationships between weather parameters and solar PV generation.
- To create a machine learning algorithm for short-term forecasting (hourly and daily) of solar energy generation to enhance load balancing and energy scheduling.
- To preprocess sensor data and build several prediction models such as linear regression, random forest, support vector regression, XGBoost, and LSTM.
- To assess the performance of prediction models based on R^2, RMSE, and MAE to select the best-performing algorithms.

SCOPE

The scope of the Short-Term Solar Energy Generation Forecasting project encompasses the building of a machine learning system which is aimed at predicting short-term power output on the basis of weather sensors data.

The following are the main scopes of the project:

Analysis of Weather and Plant Data: Study of historical data on the weather parameters (irradiance, temperature) and their influence on plant power output.

Data Preprocessing and Feature Engineering: Data cleaning and transformation for building a predictive machine learning model.

Building of Machine Learning Models: Development of several regression and time-series models – Linear Regression, Random Forest, SVR, XGBoost, and LSTM – for short-term forecasting of the plant's power generation.

Models Comparison and Evaluation: Comparison and evaluation of the developed models using such metrics as R^2, RMSE and MAE in order to find out the most effective approach to the prediction of the power output.

Model Deploying: The developed machine learning model may be integrated with the API/ Dashboard in order to accept real-time weather data and give predictions on short-term power output.

Future Model Extension: Future integration with the real-time weather API, usage of datasets from different plants and different forecasting horizons.

The project will only involve short term (hourly/daily) prediction through the use of data obtained from a single plant only, and the model will not be deployed for production.

FEASIBILITY STUDY

Feasibility study is used to determine whether the proposed project that consists of application of machine learning for short-term forecasting of solar power generation is feasible to be implemented under current resource, technological and time constraints.

1. **Technical Feasibility:** Project is technically feasible because the necessary tools for this project include open-source software libraries Python, pandas, NumPy, scikit-learn, XGBoost, TensorFlow and Keras. Dataset to be used is also open and available (Kaggle Plant 1 solar generation and weather sensor data) and preprocessing, EDA and comparison of all models have already been done and validated, and tree-based models train quickly on regular CPU.

2. **Data Feasibility:** Dataset to be used in the project is publicly available and does not require any licensing to access. Preprocessing and EDA have already been performed and it is confirmed to be clean and enough for training the models in academic scope of this project.

3**. Economic Feasibility:** The project is economically feasible because all software libraries and dataset do not cost anything. Development can be done using personal computer or free cloud services like Google Colab.

4**. Operational Feasibility:** This work is operationally feasible because the trained model could be wrapped in an API taking the weather input as an argument and forecasting the power generation output. The most effective model used (XGBoost) is also maintainable and interpretable.

5**. Schedule Feasibility:** The schedule feasibility of the project is satisfactory because the most labor-intensive part of the work (data preprocessing, exploratory data analysis, and the comparison of all models) has already been accomplished.

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PLAGIARISM REPORT

